BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: Driver Drowsiness Detection System

by

Group-05
Section-03

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Abstract

The project presents a novel Drowsiness Detection System, which is to be implemented to improve road safety by observing the key physiological and behavioral signs of driver exhaustion. The system has several sensors, such as an eye blink sensor, a gyroscope (MPU6050), an alcohol gas sensor (MQ-3), and a heart rate pulse sensor, which are connected to a microcontroller, which interprets the real-time data to activate multi-mode alerts. When the system notices signs of drowsiness, which is established by the state of eye closure, unnatural head positions (through gyroscope data), high blood alcohol content, and abnormal heart rate, the system activates a sequence of actions. These consist of auditory cues in the form of a buzzer and haptic cues in the form of a seat vibrations, and instant messages dispatched to the mobile phone of the authority with the help of an ESP-01 WiFi module. This holistic strategy is a major way of reducing the risk of accidents and death caused by driver fatigue, which is one of the major causes of commercial vehicle accidents. In addition, the system has been used as a good example of the real-life implementation of embedded systems, sensor integration, and IoT communication protocols in promoting vehicular safety programs

Keywords

Drowsiness detection, Microcontroller, Vehicle safety, Embedded system, and Sensors.

1. Introduction

Worldwide, road accidents are caused by driver fatigue, which has led to the death of many highway users in crashes. Commercial truck drivers who experience many working hours and time pressure are especially at risk. The inability to sleep may also occur because of prolonged wakefulness, which is drowsiness and involuntary. The result of these can be eye closure, nodding of the head, and lack of alertness and decision-making in serious accidents. Current drowsiness detection technologies, which are based on one sensor, such as eye tracking or steering analysis, tend to give false positives and are not very robust. There's a strong need for a whole-body real-time system that includes various physiological and behavioral measurements. This system should detect driver fatigue early enough to prevent loss of control.

Project Objectives:

1. Identify patterns of eye blink of a driver to determine unusually long eyelids that are signs of drowsiness.
2. Inertial measurements of the position and movement of the monitor head are used to detect the nodding of the head.
3. Record the heart rate variability in order to detect cardiac changes associated with fatigue.
4. Identify alcohol impairment with the aim of preventing drunk driving.

5. Create multi-mod alerts, i.e., auditory (buzzer), haptic (vibration of the seat), and digital (SMS) via (ESP-01).
6. Allow for real time remote alerts using ESP-01.

This project demonstrates fundamental concepts of computer interfacing using an interfacing of various sensors and microcontroller based hardware. It uses communication protocols such as I2C and UART and design software to successfully process more than one stream of inputs. The system depicts viable links between analog and digital sensors, the General Purpose Interface, and the Analog to Digital Converter of the microcontroller. Serial communication and wireless networking are among the main concepts in CSE360.

2. Related Work/ Inspiration

There are a few existing systems that are quite identical to our project. There is the use of computer vision to do eye tracking. This technology employs advanced systems to monitor eye position, eyelid closure, and pupil dilation. Heavy use of computer architecture is also working behind the scenes. This couldn't be implemented in our project due to the limitations of the budget and computational resources.

There are many highly accurate heartbeat sensors and analysis systems that are mostly used in the medical sector. The sensor can detect heartbeat variations more perfectly. But for our project, a moderate increase in heart rate can depict a user being in a drunk state.

Alcohol detection systems like breathalyzers or transdermal alcohol sensors are mostly used by authorized personnel. Thus, the use of such tools is out of our reach. A simple alcohol sensor is used, inspired by that.

Our project uses these three types of sensors to detect the driver's awareness status, and these are much more budget-friendly, comprehensive, and lightweight than other existing solutions in the world. Our system also works in real-time and can be adjusted to align with the admin's requirements. We also incorporate a gyro sensor, which detects the driver's head position to alert the driver if they happen to fall asleep. The combination of buzzer, vibration motor, and SMS with the ESP-01 Module all work together to guarantee the driver is alerted in times of emergency.

3. Technical Approach

☐ **System Architecture:**

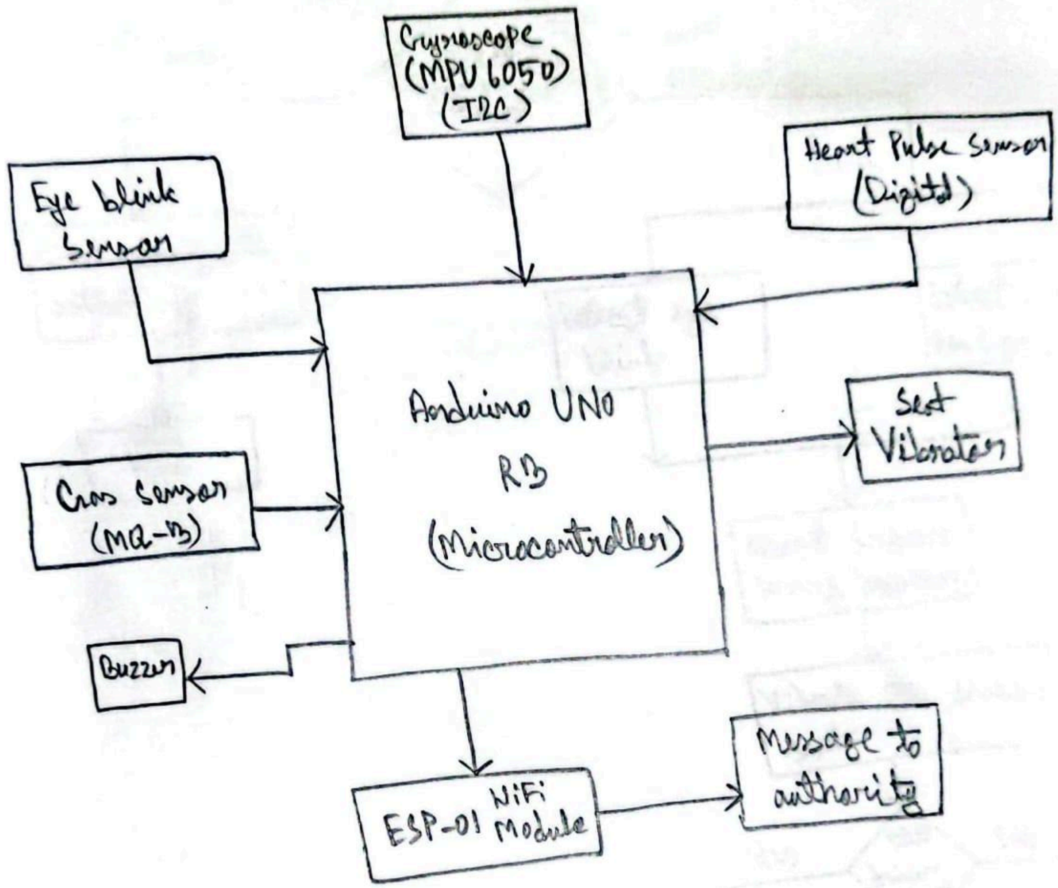
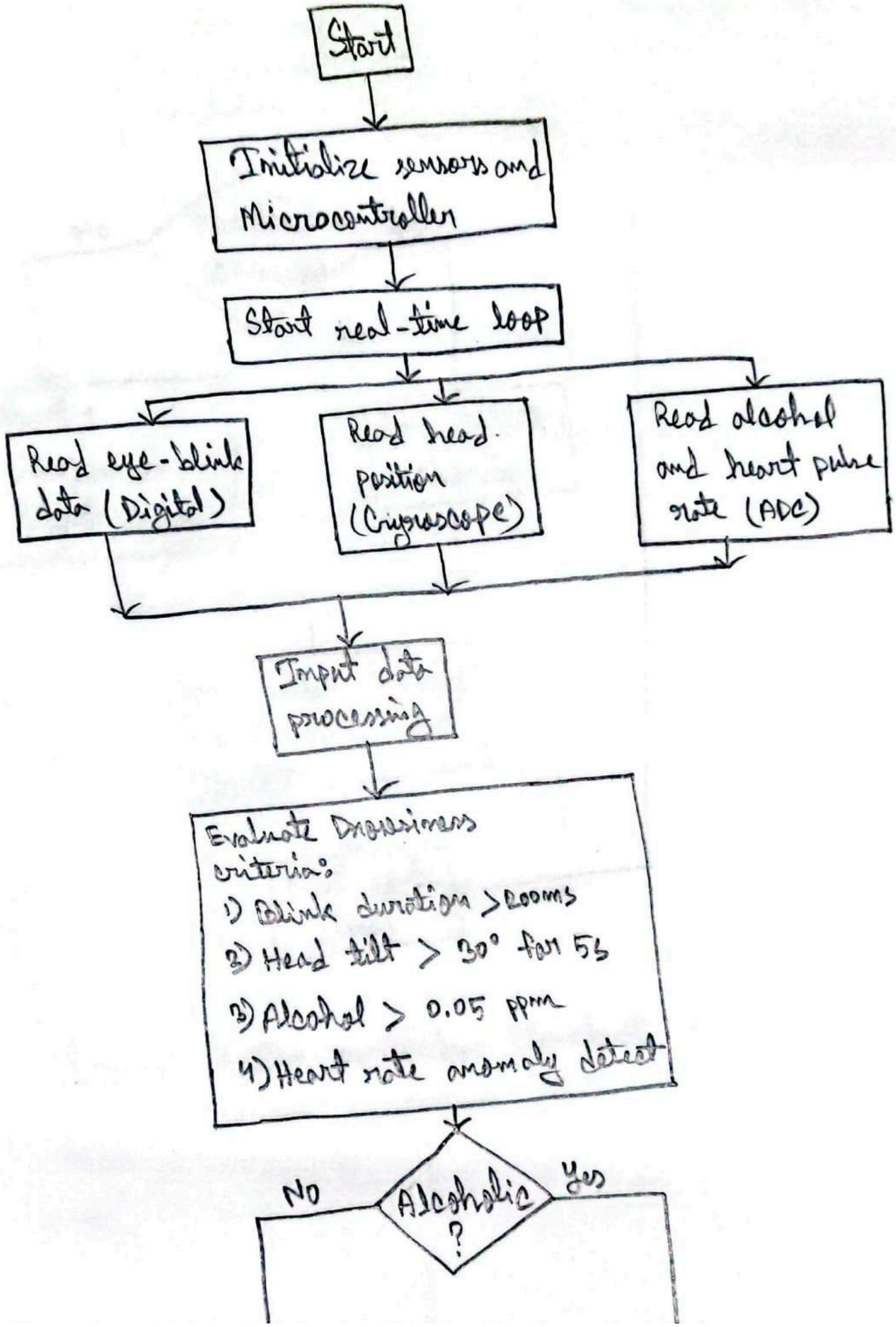


Figure: Block Diagram of Driver Drowsiness Detection System.



Start
Initialize sensors and Microcontroller
Start real-time loop
Read eye-blink data (Digital)
Read head position (Gyroscope)
Read alcohol and heart pulse rate (ADC)
Input data processing
Evaluate Drowsiness criteria:
1) Blink duration > 200ms
2) Head tilt > 30° for 5s
3) Alcohol > 0.05 ppm
4) Heart rate anomaly detect
No
Alcoholic ?
Yes

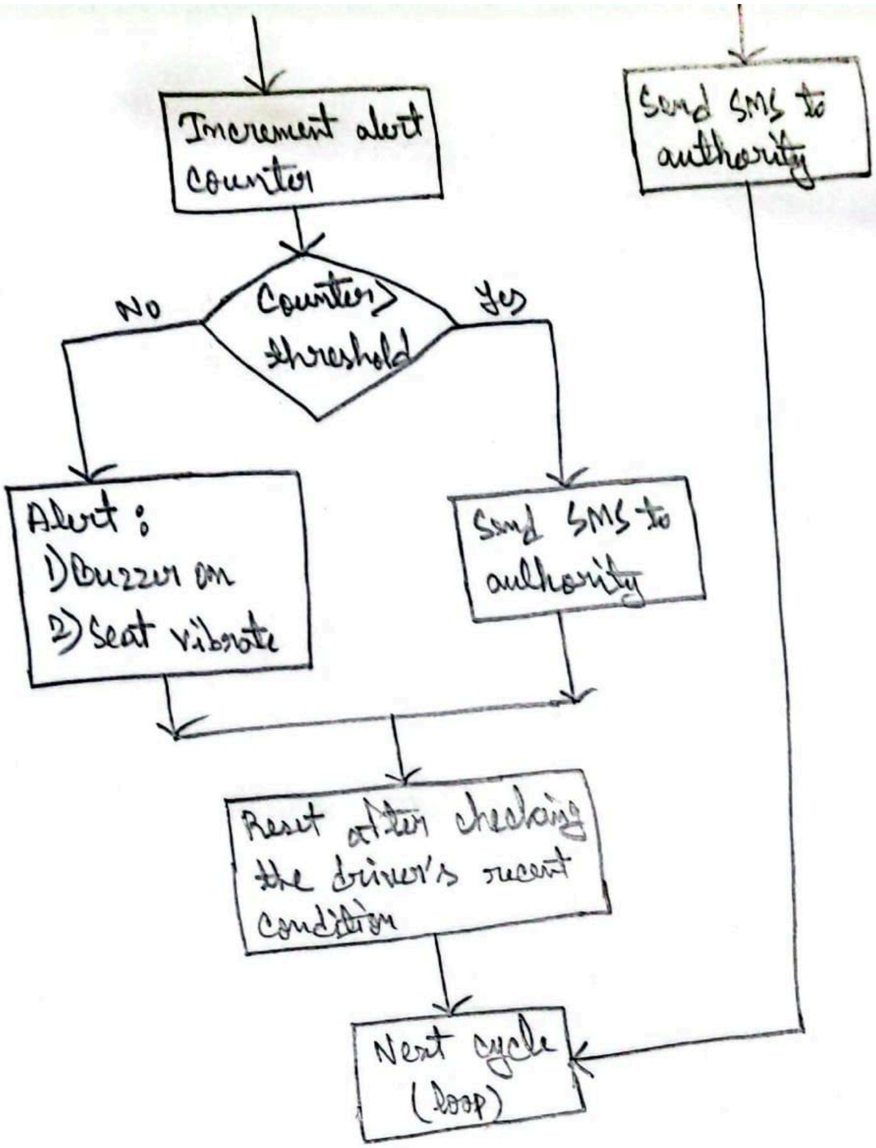


Figure: System workflow flowchart of Driver Drowsiness Detection System.

- [ ] **Components Used:**
    - **Arduino Uno R3:** The Arduino UNO R3 is a famous microcontroller. It has 14 digital pins and 6 analog pins. These pins can be used for I/O work.
    - **Arduino IDE:** The Arduino IDE is used for writing, compiling, and uploading code to the Arduino board.
    - **Buzzer:** A buzzer is an audio signaling device that is used for signaling or alerting when the given conditions are met for a project. It has a magnet which produces the sound.
    - **12V Battery:** A 12V battery is a rechargeable power source that used for powering the system.
    - **Breadboard:** A breadboard is a rectangular-shaped plastic component that is used to connect circuit elements without the use of soldering.
    - **Jumper Wires:** Jumper wires are used to connect two or multiple components.
    - **ESP-01 WIFI Module:** The ESP-01 wifi module allows the microcontroller to connect with the local wifi and send messages or signals to the devices that are on the same wifi.
    - **Heart Pulse Rate Sensor:** The heart pulse rate sensor is a device which is used to measure the pulse rate using light to sense volume changes in blood vessels.
    - **MPU 6050 Gyroscope:** The MPU6050 sensor is used to identify the movement of an object in the 6 axis (3 axis accelerometer and 3 axis gyroscope).
    - **Buck Converter:** A buck converter is used to convert the high voltage input to a lower voltage or vice versa.
    - **Eye Blink Sensor:** An eye blink sensor is used to identify the eyelid’s opening and closing situation using the infrared (IR) sensor.
    - **MQ-3 Alcohol Gas Sensor:** The alcohol gas sensor is used to identify whether a person or an environment contains alcohol or not.
    - **Shaftless Vibration Motor D34:** The D34 motor is used to give a vibrate feedback in response to an input which will be set by the user.

- [ ] **Cost Breakdown:**

Name	Price	Quantity	Total
Arduino Uno	520	1	520
Buzzer	40	2	80
9V Battery	70	1	70
Breadboard	80	1	80
JumperWires	50	2	100
ESP-01	145	1	145
Heart Rate Sensor	450	1	450
MPU6500 Gyro	165	1	165
Buck Converter	99	1	99
Eye Blink Sensor	590	1	590
Alcohol Sensor	144	1	144
Vibration Sensor	150	1	150
			Total = 2593

- ☐ **Communication Protocol:**
  - **Gyroscope Sensor (MPU6050) Using I2C:** The MPU6050 uses the I2C protocol using the synchronous communication method. It uses the I2C protocol because it is simple and suitable for low-power systems.
  - **WiFi Module (ESP-01 ESP8266) Using UART:** The ESP-01 uses the UART protocol because it is simple, efficient, and inexpensive for use of a WiFi modem for a microcontroller. To establish a communication between point-to-point applications UART is simpler than the I2C one.
  - **Alcohol Sensor (MQ-3) and Heart Pulse Sensor Using ADC:** Analog sensors make voltage outputs that are related to the heart rate and the amount of gas in the air. The microcontroller's analog-to-digital converter (ADC) then changes these outputs into digital values. This conversion helps to implement software-based threshold detection and data filtering.

☐ **Circuit/Schematic:**

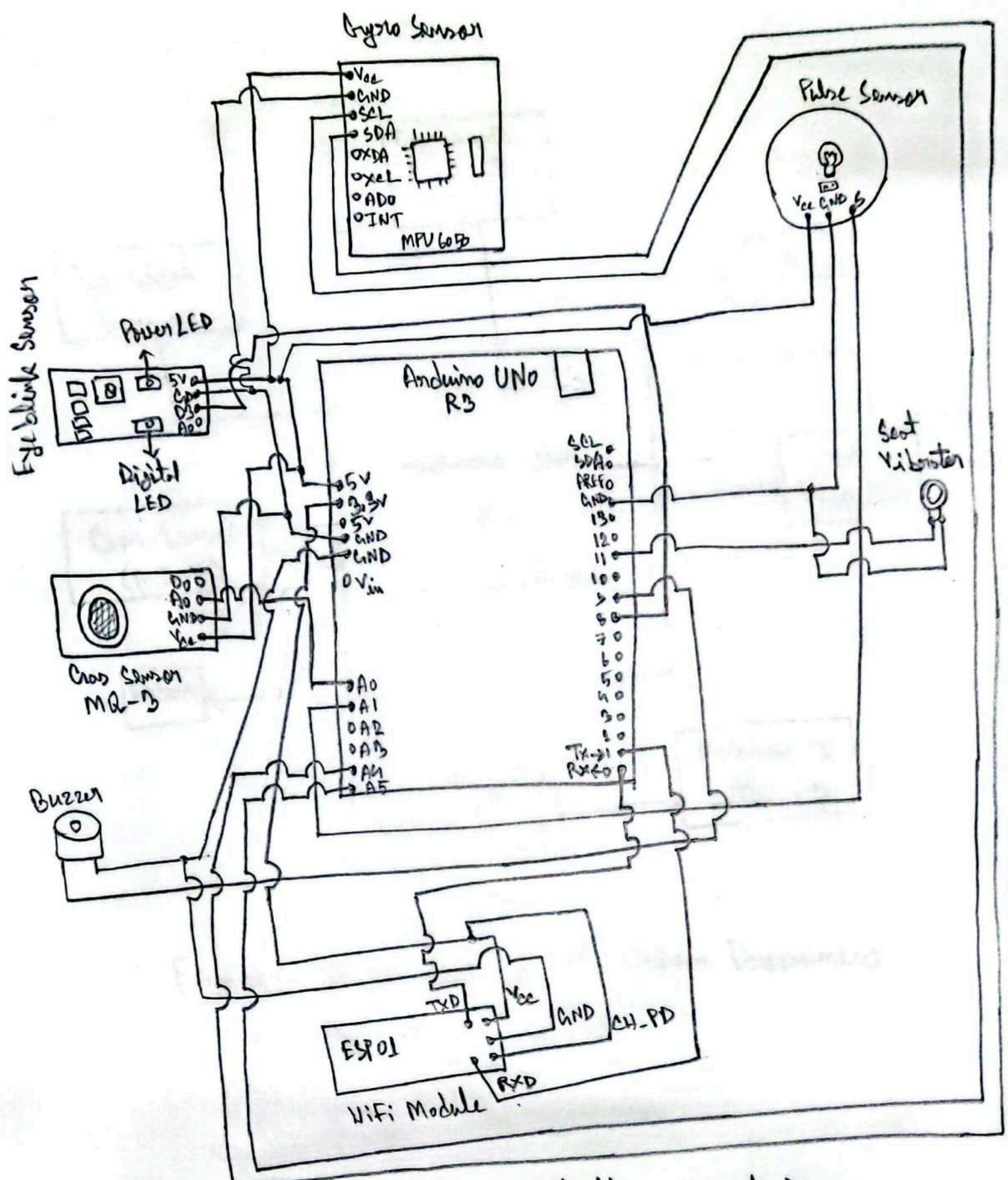


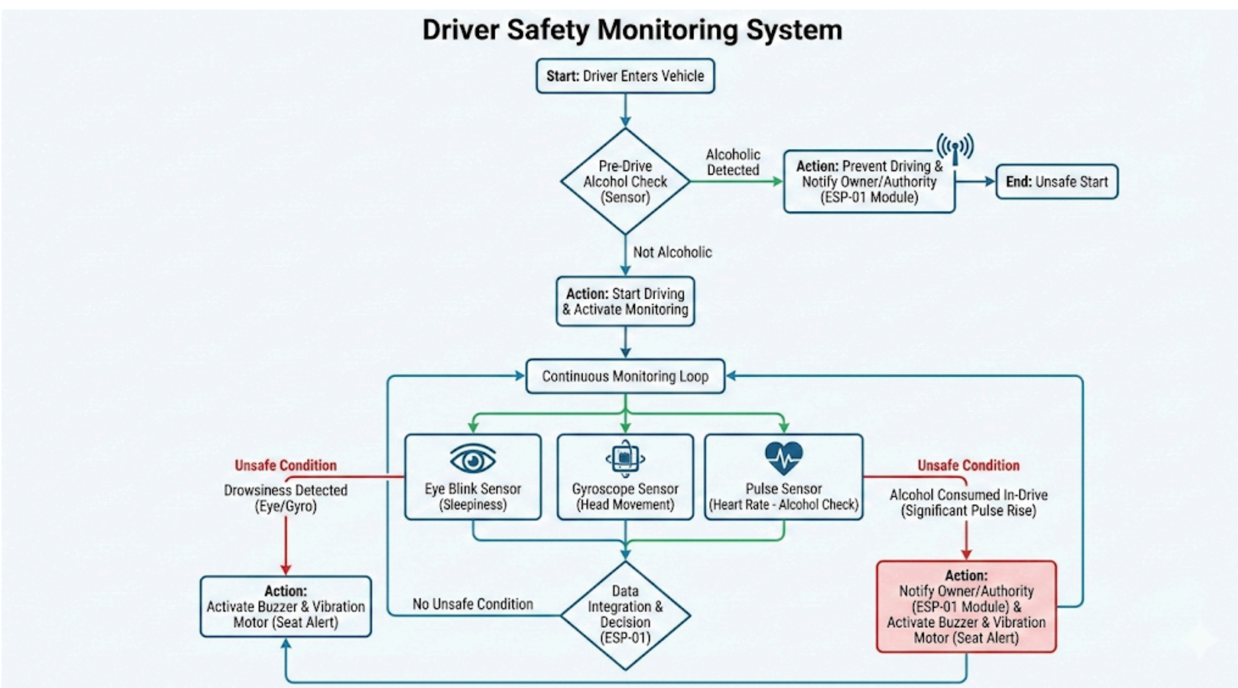
Figure: Driver Drowsiness Detection Circuit Diagram.

☐ Implementation:

Our system starts before the driver gets into the vehicle itself. It begins with checking the driver with an initial alcohol check to see whether the driver is under the influence of alcohol or not. If alcohol is detected, the system goes off, and the ESP-01 module is used to inform the vehicle owner of the situation.

Now, if the driver is not drunk, the system will allow the driver to get on the vehicle and start their journey. After that, the system will continuously check the driver's eye closure status, head position status, and their heartbeat status. If the driver is feeling sleepy and has their eyes closed consecutively for a while, the system will raise its alert. If the driver tilts their head due to feeling sleepy, the gyro sensor will detect that and again raise the alert. Moreover, if the driver decides to consume some alcohol sneakily, the pulse rate sensor will detect the spike in his heart rate and again alert the system.

The alert system consists of a buzzer that creates an alerting sound, a vibration motor that will be under the driver's seat. Once it goes off, it will vibrate the whole seat to shake up the driver. The ESP-01 module will also be used to alert the owner of the vehicle of any alerts that occur while driving.



- ☐ **Challenges:** The major challenge we faced during the project was wrong sensor readings, especially from the gyroscope and the eye blink sensor. This issue was resolved by troubleshooting with different codes and substitute components.

4. Sustainability & Impact

- ☐ **Sustainability:**
    - **Energy Efficiency:** Our system is designed to work by using a low-voltage power mode. Other components like Gyro, Heart Pulse Sensor use very low current; as a result, the system will run for a long time without impacting on fuel consumption of the vehicles.
    - **Component Lifespan & E-Waste:** Our suggested system has a modular design, so any sensor can be replaced over time. For example, if a new Alcohol Sensor needs to be upgraded, it is possible with the modularity. This can reduce electronic waste.
    - **Materials:** Our system will use recyclable as well as biodegradable materials, which ensure not only durability but also the eco-friendly standard of an alternative to industrial plastics.
- ☐ **Impact:**
    - **Safety:** Our proposed system can reduce the risk of accidents by a high margin by sensing the driver's awakeness, motion detection, and tiredness, which will save both drivers and pedestrians.
    - **Industrial & Economic Benefits:** If the accidents can be reduced by our proposed system, then logistics and transportation can be protected from damage to cargoes and fleet vehicles. For this reason, the insurance can lead to lower as well as reduced operational time caused by accidents.
    - **Driver Health Monitoring:** By using proposed sensors like the Heart Pulse Sensor, the health awareness can be optimizable. Because Commercial drivers sometimes face high stress and working conditions where if we monitor their health conditions, then we can save them from health irregularities.
- ☐ **Future Work:**
    - **GPS Integration:** As of now, our system can send SMS using the ESP module, but if a GPS system can be installed with ESP, then more accurate results can be obtained on where the vehicles are, if there are any emergency services needs that can be easily delivered.
    - **Computer Vision:** The eye blink detection can be upgraded from simple IR sensors to a camera-based system like OpenCV. This will allow for more critical analysis, like detecting abnormal eye movement using ML and Computer Vision.

- **Vehicle Interlock System:** Future implementation could be to implement directly within the vehicle system. For example, if the MQ3 sensor detects any alcohol or other system failure above the threshold value, then the system could physically prevent the engine from starting.

- [ ] **Limitations:**
    - **Environmental Interference:** In our testing phase, we saw the Eye Blink Sensor struggles with accuracy. Sometimes it was giving a false signal even if the eyes were open.
    - **Vibration Sensitivity:** The MPU6050 Gyro detects the head tilt. In our case, when we move our head frequently, it gives a heavy vibration that is extremely rough. In addition to vibration, it can sometimes create a false positive output.
    - **Connectivity:** The ESP module needs a connections to wifi network or mobile hotspot to send messages. But any chance if vehicles go to different areas, there may not be a cellular signal, which can cause of loss of signals.

5. Results & Discussion

Sensor Component	Parameter Measured	Normal Range(Safe)	Critical Threshold	Action Triggered	Performance
MQ 3 Gas Sensor	Alcohol Concentration (Analog Value 0-1023)	100 – 250	> 700	Buzzer & SMS	Take 10 - 20 ms to detect alcohol from the environment.
MPU 6050 Gyro	Head Tilt Angle (Y-axis)	-15° to +15°	> 30°	Buzzer & Seat Vibration	Need to move the head properly to sense the movement in any axis.
Eye Blink Sensor	Eye Closure Duration	< 400 ms	> 3000 ms	Buzzer & Seat Vibration	Need to close the eye to the sensor as close as possible.
Pulse Sensor	Beats Per Minute (BPM)	60 – 100 BPM	< 50 or > 120 BPM	Warning And Seat Vibration	Works accurately.

Our system results can confirm that we have met our primary objective of the project. The implementation of the eye blink sensor and MPU6050 provides us a double check mechanism for fatigue, not only relying on one but multiple to ensure the true positive results for drowsiness triggers the alarm. The vibration sensor provides an effective physical stimulation by ensuring a stronger wake-up call rather than relying only on the sound. But we saw that if the driver moves their head too much, then the Heart Pulse Sensor was very sensitive to those motion artifacts, which cause reading fluctuation sometimes. Apart from this, the core safety features like Alcohol detection can operate with a high chance of reliability for successfully detecting and identifying dangerous situations and initiating the alerts of sound and vibration to prevent potential accidents.

6. Conclusion

By integrate four sensor modalities, like eye blink, gyroscope, alcohol gas and heart rate with real-time decision logics and multiple alerts, Driver Drowsiness Detection System has successfully solved road safety issue.

Key Achievements:

1. Multiple alert systems is Digital, haptics and auditory alerts.
2. Implementation are affordable and appropriate for commercial purpose
3. Data coming in and remote monitor is made possible by ESP-01 WiFi.
4. In noise environments, reliable sensors helps to detects.

The aim of this system will be to remove sleep-induced road mishaps, which comprise more than 20 percent of the total years of death in Bangladesh (Bangladesh Road Safety Profile 2025, 2025). It has low power consumption and a recyclable design which support its environmental sustainability. The project demonstrates the principles of embedded-systems engineering that are essential to autonomous vehicles and the Internet of Things and have potential uses, such as healthcare and industrial automation. Future innovations must be involves sensor integrate and implementation to vehicle telemetry, autonomous driving architecture integration, and machine learning uses to provide custom limits. To deploy in the market, the partnership with fleet operators and compliance with the regulations is essential.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Syed Ashiqur Rahman	22301363	Alert System, Documentation, and Testing	1. Implement the alert system. 2. Write in the report 3. Testing the work of the system.
Rafi Ehtesham Mozumder	22301489	Circuit Integration, Firmware Development, Sensor Processing, and Documentation	1. Implement the circuit. 2. Write Arduino IDE code. 3. Works with various sensors and is integrated into the circuit. 4. Write in the report.
Mohd Tashwaruddin Safin	21201160	Hardware Design, Circuit Integration, Firmware Development, Sensor Processing, and Documentation	1. Design the full circuit diagram. 2. Implement the circuit. 3. Write Arduino IDE code. 4. Works with various sensors and is integrated into the circuit. 5. Write in the report.
Md Mubin Ibne Azad	22101150	Alert System, and Documentation	1. Implement the alert system. 2. Write in the report

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Bangladesh Road Safety Profile 2025. (2025). Asian Transport Observatory. Retrieved January 2, 2026, from https://asiantransportobservatory.org/analytical-outputs/roadsafetyprofiles/bangladesh-road-safety-profile-2025/

DPV Technology. (2019, September 29). *Sending data to thingspeak website using esp8266 Arduino Tutotial* [Video]. Youtube. Retrieved January 4, 2026, from https://www.youtube.com/watch?v=nMWwqcn7ofw